Computer Networks Assignment

Static Routing, Networking Devices & Random Access Protocols

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Question 1

Develop a static routing network using 4 Routers, 4 Switches and 8 PCs. The network is highlighted with its network IDs, and the IP addresses of every interface are also listed below the diagram in text format.

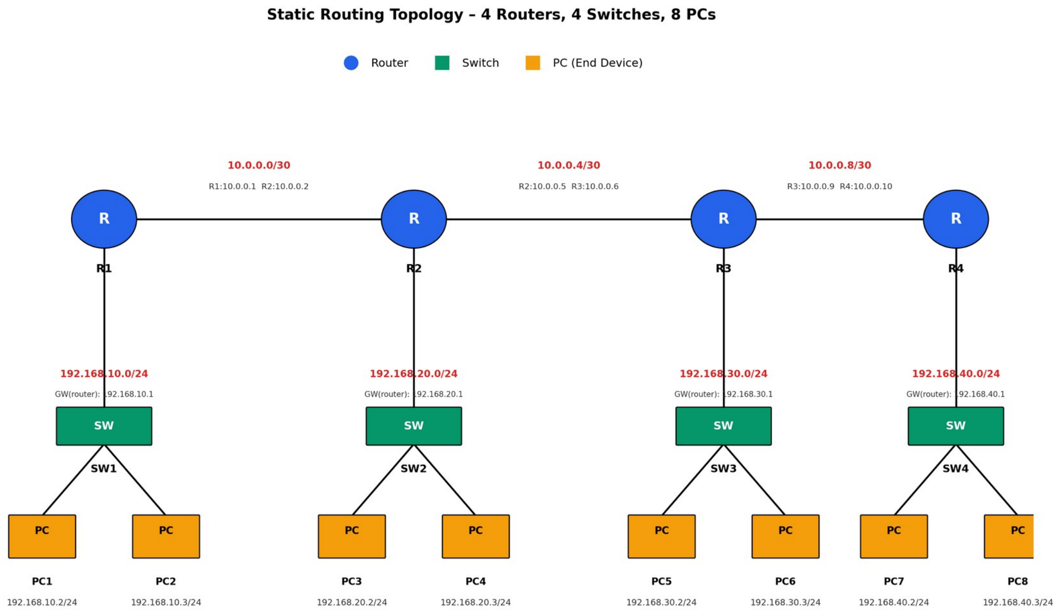

Fig 1.1 – Static routing topology (4 Routers, 4 Switches, 8 PCs)

Network IDs used in the topology

Network	Network ID	Type	Connected Devices
LAN 1	192.168.10.0/24	Ethernet (LAN)	R1 – SW1 – PC1, PC2
LAN 2	192.168.20.0/24	Ethernet (LAN)	R2 – SW2 – PC3, PC4
LAN 3	192.168.30.0/24	Ethernet (LAN)	R3 – SW3 – PC5, PC6
LAN 4	192.168.40.0/24	Ethernet (LAN)	R4 – SW4 – PC7, PC8
WAN 1	10.0.0.0/30	Serial (Point-to-Point)	R1 – R2
WAN 2	10.0.0.4/30	Serial (Point-to-Point)	R2 – R3
WAN 3	10.0.0.8/30	Serial (Point-to-Point)	R3 – R4

IP addressing table (text format)

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC1	NIC	192.168.10.2	255.255.255.0	192.168.10.1
PC2	NIC	192.168.10.3	255.255.255.0	192.168.10.1
PC3	NIC	192.168.20.2	255.255.255.0	192.168.20.1

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC4	NIC	192.168.20.3	255.255.255.0	192.168.20.1
PC5	NIC	192.168.30.2	255.255.255.0	192.168.30.1
PC6	NIC	192.168.30.3	255.255.255.0	192.168.30.1
PC7	NIC	192.168.40.2	255.255.255.0	192.168.40.1
PC8	NIC	192.168.40.3	255.255.255.0	192.168.40.1
R1	Fa0/0 (LAN)	192.168.10.1	255.255.255.0	—
R1	Se0/0/0 (to R2)	10.0.0.1	255.255.255.252	—
R2	Fa0/0 (LAN)	192.168.20.1	255.255.255.0	—
R2	Se0/0/0 (to R1)	10.0.0.2	255.255.255.252	—
R2	Se0/0/1 (to R3)	10.0.0.5	255.255.255.252	—
R3	Fa0/0 (LAN)	192.168.30.1	255.255.255.0	—
R3	Se0/0/0 (to R2)	10.0.0.6	255.255.255.252	—
R3	Se0/0/1 (to R4)	10.0.0.9	255.255.255.252	—
R4	Fa0/0 (LAN)	192.168.40.1	255.255.255.0	—
R4	Se0/0/0 (to R3)	10.0.0.10	255.255.255.252	—

Static routes required on each router

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! On R1:
ip route 192.168.20.0 255.255.255.0 10.0.0.2
ip route 192.168.30.0 255.255.255.0 10.0.0.2
ip route 192.168.40.0 255.255.255.0 10.0.0.2

! On R2:
ip route 192.168.10.0 255.255.255.0 10.0.0.1
ip route 192.168.30.0 255.255.255.0 10.0.0.6
ip route 192.168.40.0 255.255.255.0 10.0.0.6

! On R3:
ip route 192.168.10.0 255.255.255.0 10.0.0.5
ip route 192.168.20.0 255.255.255.0 10.0.0.5
ip route 192.168.40.0 255.255.255.0 10.0.0.10

! On R4:
ip route 192.168.10.0 255.255.255.0 10.0.0.9
ip route 192.168.20.0 255.255.255.0 10.0.0.9
ip route 192.168.30.0 255.255.255.0 10.0.0.9
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Question 2

Difference between Hub, Switch and Router:

Parameter	Hub	Switch	Router
OSI Layer	Physical Layer (Layer 1)	Data Link Layer (Layer 2)	Network Layer (Layer 3)
Function	Broadcasts incoming data to all ports	Forwards data only to the intended destination port using MAC address	Forwards data (routes) between different networks using IP address
Intelligence	No intelligence – pure signal repeater	Moderately intelligent – maintains a MAC address table	Highly intelligent – maintains routing tables, runs routing protocols
Collision Domain	Single collision domain for all ports	Separate collision domain per port	Separate collision domain per interface
Broadcast Domain	Single broadcast domain	Single broadcast domain (unless VLANs used)	Separate broadcast domain per interface
Addressing Used	None	MAC Address	IP Address
Speed / Efficiency	Slow, causes many collisions	Faster, reduces collisions	Comparatively slower per-packet processing due to routing decisions, but connects entire networks efficiently
Typical Use	Legacy LANs (obsolete today)	Connecting devices within the same LAN	Connecting different networks / LAN to WAN (e.g., Internet)

Why the Router is the most intelligent device: A router operates at the Network Layer and makes forwarding decisions based on logical (IP) addressing rather than just physical signals. It builds and maintains routing tables, can run dynamic routing protocols (RIP, OSPF, EIGRP, etc.) to learn the best path, performs packet filtering, NAT, and can connect entirely different types of networks (e.g., LAN to WAN). This requires it to understand network topology and make intelligent path-selection decisions — something neither a hub nor a switch is capable of.

Why the Hub is the least intelligent device: A hub works purely at the Physical Layer. It has no knowledge of MAC or IP addresses and simply regenerates and repeats every incoming electrical signal out to all connected ports except the one it arrived on, with no filtering or decision-making of any kind. This causes unnecessary traffic, increases collisions, and wastes bandwidth, which is why hubs are now obsolete and have been replaced by switches.

Question 3

What are Random Access Protocols?

Random Access Protocols (also called Contention-based protocols) are a category of Multiple Access Control (MAC) protocols in which no station is superior to another, and none is assigned control over another. Every station has the right to access the shared communication medium at any time without needing prior permission from a central controller. Because more than one station may attempt to transmit at the same instant, collisions can occur, so each protocol defines a set of rules that specify what a station should do when the medium is free, and how to detect and recover from a collision if one occurs.

Why do we need Random Access Protocols?

- In a shared broadcast medium (e.g. a LAN or wireless channel), several stations compete for the same channel; a mechanism is required to decide who transmits and when.
- Fixed allocation schemes (like TDMA/FDMA) waste bandwidth when a station has no data to send, since the slot/frequency stays reserved for it.
- Random access protocols make efficient use of channel capacity by allowing any idle station to transmit as soon as it has data, improving throughput under bursty, unpredictable traffic.
- They avoid the need for a centralized controller, making the network simpler, cheaper and more scalable.

Categories of Random Access Protocols

1. ALOHA

Developed at the University of Hawaii, ALOHA allows a station to transmit whenever it has data, without checking whether the channel is free. If two frames collide, both are destroyed and must be retransmitted after a random back-off delay.

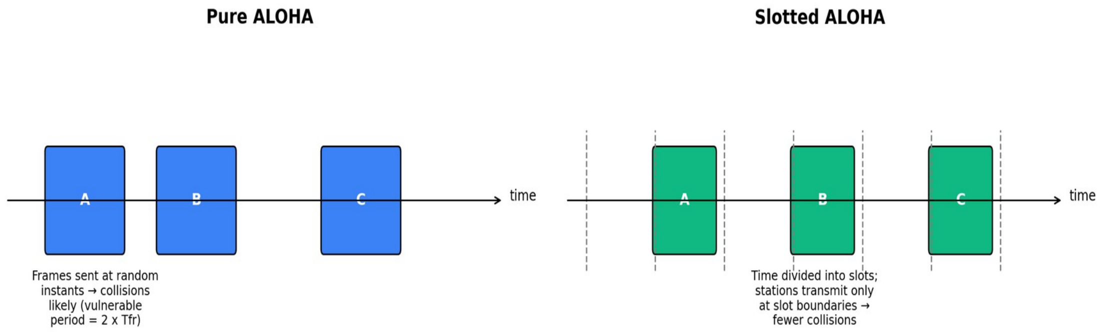

Fig 3.1 – Pure ALOHA vs Slotted ALOHA

- Pure ALOHA: A station transmits immediately whenever a frame is ready. Collisions are highly likely because there is no time synchronisation; the vulnerable time is 2 x Tfr (twice the frame transmission time). Maximum channel utilisation ≈ 18.4%.
- Slotted ALOHA: Time is divided into discrete slots equal to the frame transmission time, and a station may only start transmission at the beginning of a slot. This halves the vulnerable time to Tfr and raises maximum utilisation to ≈ 36.8%.

2. CSMA (Carrier Sense Multiple Access)

CSMA improves on ALOHA by requiring every station to “listen before talking” — i.e., sense the carrier/channel for existing transmissions before sending its own frame. This reduces, but does not eliminate, collisions (since propagation delay means two far-apart stations may both sense an idle channel and transmit at almost the same time).

- 1-Persistent CSMA: If the channel is busy, the station keeps sensing continuously and transmits as soon as it becomes idle (high chance of collision if multiple stations are waiting).
- Non-Persistent CSMA: If the channel is busy, the station waits a random amount of time before sensing again, reducing collisions but increasing delay.

- P-Persistent CSMA: Used in slotted channels; if idle, the station transmits with probability p, or waits for the next slot with probability (1–p) — balances the trade-off between the previous two schemes.

3. CSMA/CD (Collision Detection)

Used mainly in traditional wired Ethernet (IEEE 802.3). While transmitting, the station continues to monitor the channel; if a collision is detected (by sensing abnormal voltage levels), it immediately stops transmission, sends a short jam signal to inform all stations of the collision, and then waits for a random back-off time (binary exponential back-off) before retrying.

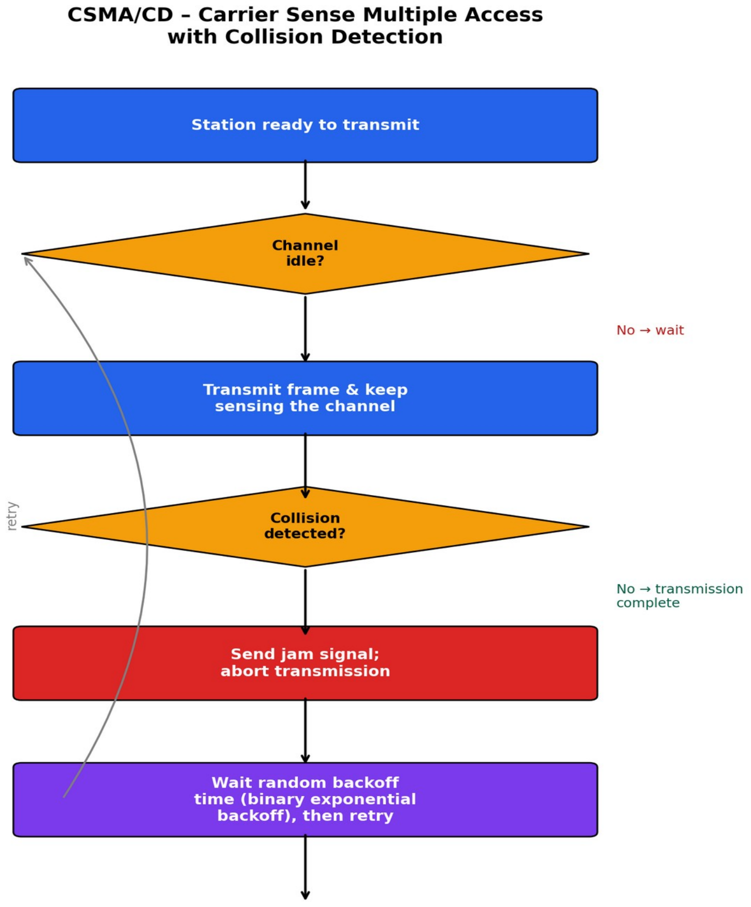

Fig 3.2 – CSMA/CD operation flow

4. CSMA/CA (Collision Avoidance)

Used mainly in wireless LANs (IEEE 802.11), where collisions cannot be directly detected by the sender (because a transmitting station cannot simultaneously listen for collisions on a wireless medium, and hidden-node problems exist). Instead, CSMA/CA tries to avoid collisions altogether using techniques such as Interframe Space (IFS) delays, a random back-off timer, and an RTS/CTS (Request-to-Send / Clear-to-Send) handshake, followed by an acknowledgement (ACK) for every successfully received frame.

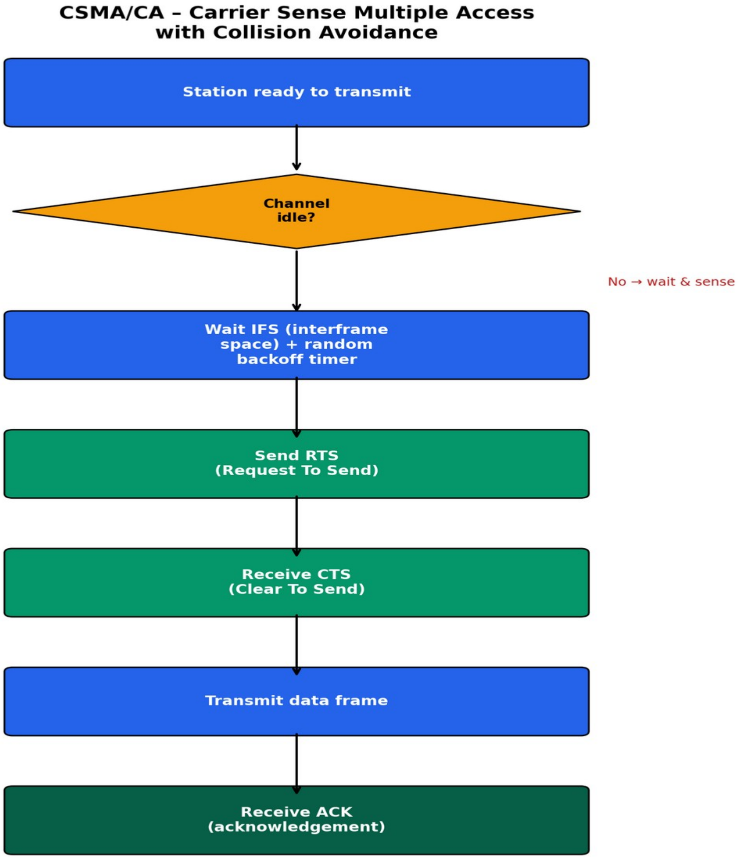

Fig 3.3 – CSMA/CA operation flow

Difference between CSMA/CD and CSMA/CA

Parameter	CSMA/CD	CSMA/CA
Full form	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Used in	Wired networks (traditional Ethernet, IEEE 802.3)	Wireless networks (Wi-Fi, IEEE 802.11)
Approach	Detects a collision after it occurs, then stops and retransmits	Tries to avoid collisions before they occur
Collision handling	Sends a jam signal and aborts transmission on detecting a collision	Uses RTS/CTS handshake and random back-off to minimise the chance of collision
Acknowledgement	Not mandatory – collision itself indicates failure	Positive ACK required for every frame since collisions can't be directly sensed
Medium characteristic	Transmitting station can sense the medium while sending	Transmitting station cannot reliably sense collisions while transmitting
Efficiency	Efficient for wired shared medium	Extra overhead (RTS/CTS/ACK) reduces throughput but is necessary for wireless reliability